

Problem F. 50 Years of Excellence

Input file: standard input
Output file: standard output
Time limit: 1 second
Memory limit: 1024 megabytes

The International Collegiate Programming Contest has inspired generations of students throughout fifty years of excellence.

Consider n consecutive years. In each year, exactly m problems are prepared, and every problem has an integer rating.

The score of a year is the sum of the ratings of all m problems in that year. The score immediately before the first year is defined to be 0.

A year is called *excellent* if its score is strictly less than the score of the previous year. This definition also applies to the first year, whose score is compared with 0.

Determine how many years are excellent.

Input

The first line contains two integers n and m ($1 \leq n \leq 50$, $1 \leq m \leq 13$)—the number of years and the number of problems in each year. Each of the next n lines contains m integers $a_{i,1}, a_{i,2}, \dots, a_{i,m}$ ($-1000 \leq a_{i,j} < 0$)—the ratings of the problems in year i .

Output

Print one integer—the number of excellent years.

Examples

standard input	standard output
5 3 -100 -200 -300 -100 -200 -250 -150 -200 -200 -100 -150 -250 -200 -200 -250	2
4 2 -5 -5 -6 -5 -10 -1 -6 -6	3

Note

In the first example, the scores of the five years are -600 , -550 , -550 , -500 , and -650 . The first year is excellent because $-600 < 0$, and the fifth year is excellent because $-650 < -500$. Equality is not sufficient, so the third year is not excellent.

In the second example, the scores are -10 , -11 , -11 , and -12 . The first, second, and fourth years are excellent.